# AI in Daily Life

### Duration

6 hours of detailed lecture content — 30 minutes per topic

### Objective

Understand practical AI applications in everyday scenarios with real-world examples and ethical grounding

01

### Hour 01

AI for Shopping & Personal Finance

02

### Hour 02

AI Tools and Frameworks

03

### Hour 03

AI in Healthcare and Robotics

04

### Hour 04

Understanding Bias in AI

05

### Hour 05

Robotics and AI Integration

06

### Hour 06

Comprehensive Review & Practical Applications

HOUR 01 — 30 MINUTES

# AI for Shopping & Personal Finance

**Definition:** AI shopping refers to using artificial intelligence technologies to enhance the shopping experience, automate purchasing decisions, and optimize personal financial management. It transforms your smartphone into a digital personal assistant that handles research, price tracking, and checkout processes.

## Key Applications in Shopping

### Personalized Recommendations

AI analyzes your browsing history, purchase patterns, and preferences to suggest products you're likely to buy. Amazon's recommendation engine suggests items based on items you've viewed and purchased before, **increasing sales by up to 35%.**

### Price Tracking & Comparison

AI tools monitor price fluctuations across multiple retailers in real-time. The system alerts you when the price drops below your target price. This can **save consumers 15–30%** on purchases.

### Agentic Commerce

Advanced AI agents can autonomously navigate retail websites, find the best prices, compare specifications, and even complete checkout on your behalf with your permission. This is the next frontier in AI shopping.

### Visual Search

Upload a photo of an item you like, and AI finds similar products available for purchase. Particularly useful for fashion and home décor shopping.

## AI in Personal Finance

According to **Experian's 2024 report**, 67% of Gen Z and 62% of millennials use AI for personal finance tasks, with **98% reporting positive experiences.**

### Budgeting & Savings

AI chatbots like ChatGPT help create personalized budgets by analyzing your spending patterns. They identify areas where you overspend and suggest realistic savings targets. Example: An AI system might notice you spend **$200/month on subscriptions** and recommend canceling unused services.

### Investment Planning

AI tools analyze market trends and your risk tolerance to suggest investment strategies. **48% of users** report using AI for investment planning decisions.

### Credit Score Improvement

AI provides actionable steps to improve credit scores — analyzing your credit report and recommending specific actions like paying down high-interest debt or disputing errors. **48% of users** have used AI for this purpose.

### Expense Tracking

AI automatically categorizes transactions and identifies spending patterns. It can alert you when you're approaching budget limits in specific categories.

### Benefits

- **Time savings:** Users report 5–10 hours saved monthly
- **Cost reduction:** 15–30% savings on purchases
- **Accessibility:** Free tools widely available
- **Simplification:** Complex financial tasks made manageable

**Important Risks:** AI can hallucinate or provide inaccurate information. Financial advisors emphasize that AI should be a starting point, not your only resource. Always verify important financial decisions with human experts. AI systems may have biases in recommendations based on their training data.

## Practical Tips for Safe AI Shopping

### Verify Before You Buy

Always verify AI recommendations with multiple sources before making large purchases.

### Use Budgets as Guidelines

AI-generated budgets are helpful guides, not absolute rules — adjust based on your personal circumstances.

### Set Transaction Limits

Enable transaction limits on AI shopping agents to prevent large unauthorized purchases.

### Seek Professional Advice

Review AI recommendations for investment advice with a certified financial advisor. AI assists — it does not replace — professional guidance.

HOUR 02 — 30 MINUTES

# AI Tools and Frameworks

**Definition:** AI tools and frameworks are software libraries and platforms that provide pre-built functions, algorithms, and models to help developers build AI applications without starting from scratch. They abstract complex mathematical operations into simple, usable functions.

## Why Do We Need Frameworks?

### Complexity Reduction

Building AI from scratch requires deep knowledge of linear algebra, calculus, and statistics. Frameworks handle this complexity, allowing developers to focus on solving problems.

### Speed to Market

Pre-built components accelerate development. What might take months to build from scratch can be accomplished in weeks using frameworks.

### Community Support

Popular frameworks have large communities providing tutorials, solutions to common problems, and continuous improvements.

### Optimization

Frameworks are optimized for performance on different hardware — CPUs, GPUs, and specialized processors.

## Major AI Frameworks for Beginners

| Framework | Primary Use | Real-World Example | Best For |
|---|---|---|---|
| **Scikit-learn** | Traditional machine learning | A bank classifies loan applications as "approve" or "deny" using applicant income, credit score, and employment history | Beginners; excellent documentation |
| **PyTorch** | Deep learning & neural networks | Meta uses PyTorch to power recommendation systems on Facebook and Instagram | Research; flexible & intuitive design |
| **TensorFlow** | Deep learning, production deployment | Google Translate — translating text across 100+ languages | Production systems; mobile & edge deployment |
| **FastAI** | Simplified deep learning | A farmer builds a model that identifies crop diseases from photos in just 5 lines of code | Fastest entry point to deep learning |
| **PyCaret** | Automated Machine Learning (AutoML) | A startup builds a customer churn prediction model without hiring specialized data scientists | Rapid prototyping; 20 lines of code to deployment |

## How to Choose a Framework

1

### Beginners

Start with **scikit-learn** for traditional ML or **FastAI** for deep learning

2

### Rapid Prototyping

Use **PyCaret** to quickly test ideas and compare multiple models

3

### Research

**PyTorch** is the preferred choice in academia

4

### Production

Use **TensorFlow** or **PyTorch** depending on your deployment needs

## Core Concepts Every Trainee Must Know

- Training Data
- Model
- Prediction
- Accuracy

HOUR 03 — 30 MINUTES

# AI in Healthcare and Robotics

**Overview:** AI is revolutionizing healthcare by improving diagnosis accuracy, accelerating drug discovery, and enabling personalized treatment plans. The global healthcare AI market is growing at **40% annually.**

### Diagnostic AI: Key Stats

**94.5%**

AI Mammogram Accuracy

vs. 88% for human radiologists

**40%**

Annual Market Growth

Global healthcare AI market expansion

**1M+**

Hospital Deliveries

Moxi robots across 25+ hospitals

### Diagnostic AI Systems

- **Medical Imaging Analysis:** AI algorithms analyze X-rays, MRIs, and CT scans to detect diseases like cancer, pneumonia, and heart conditions. Studies show AI can match or exceed radiologist accuracy in detecting certain cancers.
- **Pathology Analysis:** AI examines tissue samples to identify diseases — faster and more consistent than manual analysis by pathologists.
- **Symptom Checkers:** AI chatbots ask targeted questions about symptoms and suggest possible conditions. Important: these guide you, they do not replace doctors.

## Robotic Surgery & Assistance

### Surgical Robots (da Vinci System)

The surgeon controls the robot from a console, and the robot translates their movements into precise surgical actions. Benefits: smaller incisions, less blood loss, faster recovery times.

### Autonomous Ultrasound (2026)

Robotic ultrasound systems with AI can autonomously locate tumors with an **84% success rate in 5.75 seconds**. AI identifies the target; geometric algorithms position the probe correctly.

### Hospital Logistics — Moxi Robot

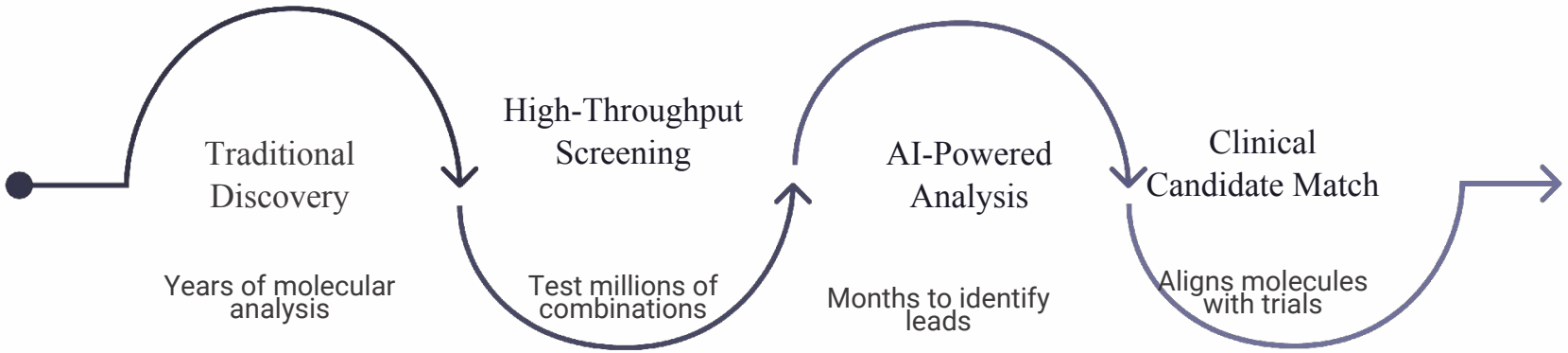
Moxi from Diligent Robotics transports medications, lab samples, and supplies between departments. Over **1 million deliveries in 25+ hospitals** — returning hundreds of thousands of hours to clinicians for patient care.

### Therapeutic & Emotional Support

**Robin** — a 4-foot-tall robot that sounds like a 7-year-old girl — is deployed in children's hospitals to reduce fear and loneliness. It remembers individual patients and asks about their progress.

HOUR 03 — 30 MINUTES

# AI for Drug Discovery




AI analyzes millions of molecular combinations to identify promising drug candidates. A process that traditionally took years now takes months. AI also identifies patients who are strong candidates for specific clinical trials, accelerating research timelines significantly.

## Important Limitations & Ethical Considerations

**AI is a Tool, Not a Replacement**

AI assists doctors but never replaces human judgment. Doctors must understand and approve all AI recommendations before acting on them.

**Data Privacy**

Healthcare AI requires sensitive patient data. Strict regulations — including HIPAA in the US — protect this data and govern its use.

**Bias in Healthcare AI**

If training data comes from biased sources, AI can perpetuate healthcare disparities. This is explored in depth in Hour 04.

**Regulatory Approval**

Healthcare AI systems must undergo rigorous testing and regulatory approval before any clinical use. No shortcuts are acceptable in this domain.

HOUR 04 — 30 MINUTES

# Understanding Bias in AI

**Definition:** Bias in AI refers to systematic errors or unfair outcomes that AI systems produce, often disadvantaging certain groups of people. Unlike human bias — which is individual — **AI bias is systematic and affects many people consistently.**

## The Four Types of AI Bias

### 1. Historical / Representational

Training data reflects historical inequalities and discrimination.

*Example: A hiring AI trained on historical data learns to discriminate against women because fewer women were hired in tech roles historically — and perpetuates this pattern.*

### 2. Selection / Measurement

The data collected is incomplete or doesn't accurately measure what we want to measure.

*Example: A healthcare algorithm trained primarily on data from wealthy hospitals may not work well for patients from underserved communities.*

### 3. Algorithmic / Optimization

The algorithm itself — even with unbiased data — can produce biased outcomes.

*Example: An algorithm optimizing for "profit" in lending might systematically deny loans to certain neighborhoods, even if individual applicants are qualified.*

### 4. Feedback / Emergent

Bias emerges over time as the system learns from its own decisions.

*Example: A criminal justice AI recommends more police patrols in certain neighborhoods → more arrests → reinforces the original bias. A vicious cycle.*

## Real-World Case Studies

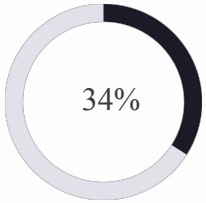
### Case Study 1: Healthcare Algorithm Racial Bias (2019)

A widely-used algorithm identified patients needing extra care but showed **significant racial bias**. At the same risk score, Black patients were considerably sicker than White patients. Why? The algorithm predicted *healthcare costs*, not actual health needs. Because of unequal access to care, less money is spent on Black patients with the same health needs — so the algorithm falsely concluded they were healthier.

Fixing this bias would increase Black patients receiving additional help from **17.7% to 46.5%** — a life-changing difference.

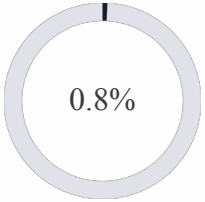
### Case Study 2: Facial Recognition Bias

Facial recognition systems trained primarily on light-skinned faces show dramatically unequal error rates:




Error Rate

Dark-skinned women




Error Rate

Light-skinned men

This bias has led to **wrongful arrests of innocent people**, disproportionately affecting Black men.

## Why Bias Matters & How to Fix It

### Why It Matters

- **Individual Impact:** People are denied jobs, loans, healthcare, and opportunities based on biased AI decisions
- **Systemic Impact:** Biased AI amplifies existing inequalities across society
- **Legal Impact:** Biased AI can violate anti-discrimination laws
- **Trust Impact:** Biased AI erodes public confidence in AI systems

### Mitigation Strategies

- **Diverse Training Data:** Ensure training data represents all demographic groups
- **Fairness-Aware Modeling:** Test models separately for each demographic group
- **Bias Auditing:** Regularly test deployed AI systems for bias
- **Transparency:** If an AI denies you a loan, you should be able to understand why
- **Human Oversight:** AI should assist, not replace, human judgment in high-stakes decisions
- **Diverse Teams:** Diverse perspectives catch biases that homogeneous teams miss

Bias in AI is not a technical problem alone — it's a **socio-technical problem** requiring diverse teams, diverse data, diverse perspectives, and ongoing vigilance. As AI becomes more powerful, addressing bias becomes more critical.

HOUR 05 — 30 MINUTES

# Robotics and AI Integration

**Definition:** Robotics is the field of engineering and computer science that designs, builds, and programs machines (robots) to perform tasks autonomously or semi-autonomously. When combined with AI, robots can perceive their environment, make decisions, and act intelligently.

## The Three Pillars of Intelligent Robots

### 1. Perception (Input)

The robot's ability to sense and understand its environment using cameras, LiDAR, ultrasonic sensors, thermal cameras, and microphones. Deep learning processes sensor data to identify objects and detect obstacles. *Challenge: Real-world environments are messy and unpredictable.*

### 2. Cognition (Decision-Making)

The robot's ability to understand the situation and decide what to do. Large Language Models (LLMs) enable robots to understand natural language commands. Example: "Pick up the red cup and place it on the table" → broken into sub-tasks: locate, grasp, move, release. *Challenge: Handling ambiguity and unexpected situations.*

### 3. Control (Output/Action)

The robot's ability to execute actions in the physical world via motors, actuators, and mechanical systems. AI learns optimal movement patterns — for example, grasping objects of different shapes without dropping them. *Challenge: Physical systems have inertia, friction, and unpredictability.*

## Types of Robots & Real-World Examples

### Mobile Robots

**Purpose:** Navigate and move through environments — delivery robots, warehouse robots, autonomous vehicles, quadrupeds.

**AI Challenge:** Localization, mapping, path planning, and obstacle avoidance.

**Example:** Boston Dynamics' Spot navigates complex terrain, climbs stairs, and opens doors using AI-powered perception and control.

### Manipulator Robots

**Purpose:** Perform precise tasks with arms and hands — industrial robots, surgical robots, humanoid robots.

**AI Challenge:** Grasping objects without breaking them, dexterity, and task planning.

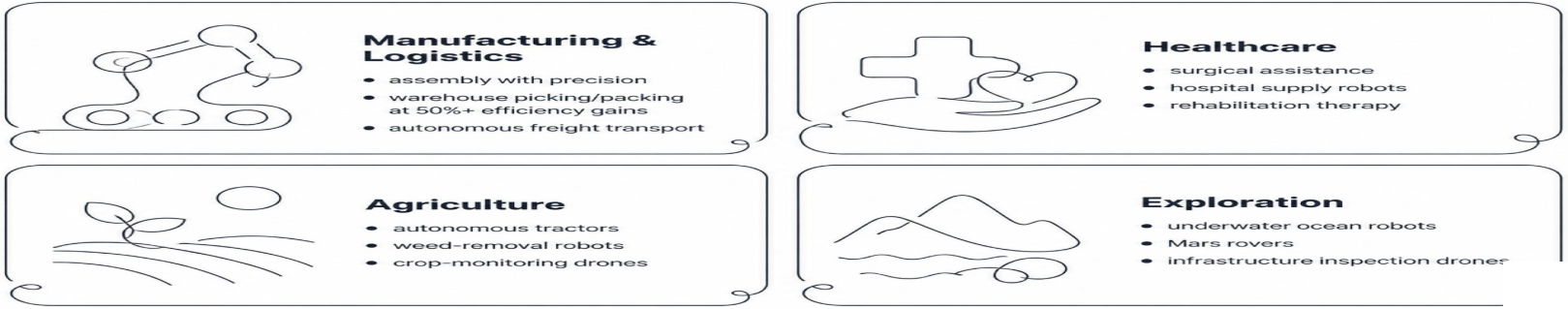
**Example:** Gemini Robotics (2026) can fold origami, play cards, and perform complex manipulation tasks by learning from demonstrations.

### Humanoid Robots

**Purpose:** Mimic human form and capabilities — Tesla Bot, Boston Dynamics Atlas, Honda Asimo.

**AI Challenge:** Balance, coordination, and understanding human environments designed for humans.

## Applications Across Industries



# Current Challenges & The Future

Current limitations and future directions for robotics AI.

## Current Challenges

**Sim-to-Real Gap**: Robots trained in simulation often fail in the real world due to unpredictable factors
**Generalization**: A robot trained to grasp red cups may fail with blue cups
Safety: Unexpected robot behavior near humans could cause injury
Cost: Advanced robots are expensive, limiting adoption
Fragility: Current AI-controlled robots are sensitive to prompt specification and exhibit non-deterministic behavior

## The Future of Robotics AI

**Generalist Robots**: Future robots will learn new tasks quickly rather than being trained for one specific task
**Human-Robot Collaboration**: Robots handle physical tasks; humans provide judgment and oversight
**Embodied AI:** Models like Gemini Robotics enable robots to interact with the physical world more naturally
Swarm Robotics: Multiple robots coordinating to accomplish complex tasks

# Comprehensive Review & Practical Applications

This final session connects all five hours of content, reinforces core concepts, and applies them through practical exercises and ethical reflection.

## How the Topics Connect




## Core Concepts Glossary

### Machine Learning

Teaching computers to learn from data rather than explicitly programming them. The computer identifies patterns in data and uses those patterns to make predictions.

### Deep Learning

A subset of machine learning using neural networks inspired by the human brain. Excellent for complex tasks like image recognition and natural language understanding.

# Core Concepts & Practical Exercises

Core concepts followed by applied exercises.

### Training

The process of feeding data to an AI model so it learns patterns. Requires quality data, appropriate algorithms, and careful tuning.

### Inference

Using a trained model to make predictions on new data. This is what happens when you actually use an AI system in practice.

## Practical Exercises

### Exercise 1: Identify the Bias

A company uses AI to screen job applications. The AI rejects 80% of applications from women but only 20% from men — even though women applicants have similar qualifications. What type of bias is this? How would you fix it?

Answer: This is historical/representational bias. The training data likely came from a company with a history of hiring more men. Fix: audit training data for gender representation, retrain with balanced data, and test the model separately for men and women to ensure equal performance.

### Exercise 2: Choose the Right Tool

You want to predict house prices based on features like size, location, and age. Which framework would you use? Why?

Answer: Scikit-learn is appropriate. This is a regression problem (predicting a continuous value) with structured data. Scikit-learn has excellent regression algorithms and is easy to learn for beginners.

# Exercise 3: Identify AI in Your Daily Life

List 5 AI applications you use. For each, identify: (a) What data does it use? (b) What decision does it make? (c) What could go wrong?

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| Application | Data Used | Decision Made | What Could Go Wrong |
|---|---|---|---|
| Netflix Recommendations | Viewing history & ratings | Predicts what you'll enjoy watching | Could recommend inappropriate content |
| Google Maps | Location, speed, traffic data | Predicts travel time & optimal route | Could be inaccurate during unusual events |
| Email Spam Filter | Email content & sender history | Classifies as spam or not spam | Could filter legitimate emails (false positive) |

## Exercise 3 Prompt

List 5 AI applications you use. For each, identify: (a) What data does it use? (b) What decision does it make? (c) What could go wrong?

# Ethical Principles for AI Professionals

Core principles for responsible AI practice.

### Transparency

Users should understand how AI systems make decisions that affect them. Black-box decisions in high-stakes domains are unacceptable.

### Fairness

AI systems should treat all groups equitably. Test models across demographic groups before deployment.

### Accountability

Someone must be responsible when AI systems cause harm. Responsibility cannot be diffused away by complexity.

### Privacy

User data must be protected and used only for stated purposes. Consent and data minimization are essential.

# Future Trends to Watch

Emerging directions highlighted in the course.

| | |
|---|---|
| **Multimodal AI** | Systems that understand text, images, audio, and video together — e.g., Gemini analyzing a photo and answering questions about it |
| **Agentic AI** | AI that autonomously plans and executes complex tasks — e.g., shopping agents completing entire purchases without human intervention |
| **Embodied AI** | AI designed specifically for robots and physical interaction with the world — Gemini Robotics as a leading example |
| **Explainable AI** | Systems that explain their decisions in human-understandable terms — critical for trust in high-stakes domains |
| **Federated Learning** | Training AI models on distributed data without centralizing sensitive information — privacy-preserving by design |

AI is a powerful tool — not magic. It requires quality data, thoughtful design, and human oversight. The most successful AI professionals combine technical skills with ethical awareness and a commitment to building AI systems that benefit everyone, not just a privileged few.